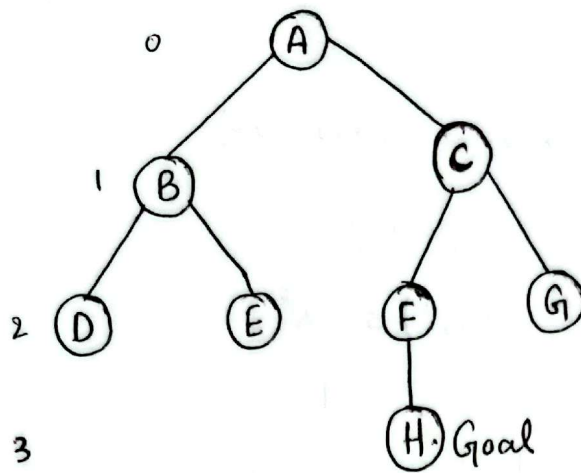


Q1. Breadth-First Search



A			
B	C		
D	E		
D	E	F	G
F	G		
F	G		
G	H		
H			
empty			

Time Complexity

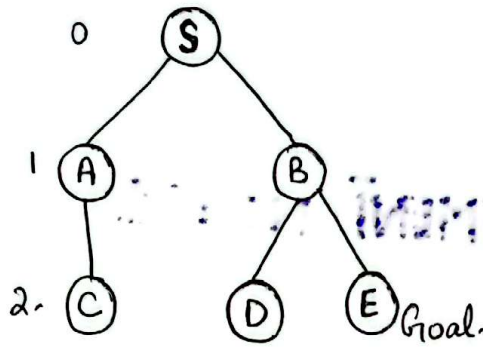
$$O(b^d) = 2^3 = 8$$

Completeness: Complete

Path

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H$

Q2. Breadth-First Search



S	
A	B
B	C
C	D
D	E
E	
empty	

Path:

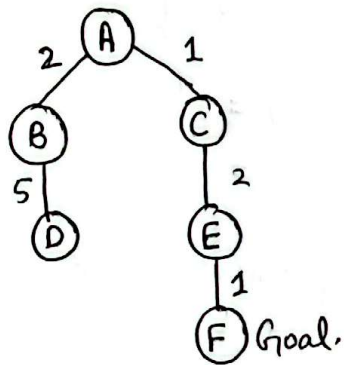
$S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

Time Complexity

$$O(b^d) = 2^2 = 4$$

Completeness: Complete

Q3. Uniform Cost Search



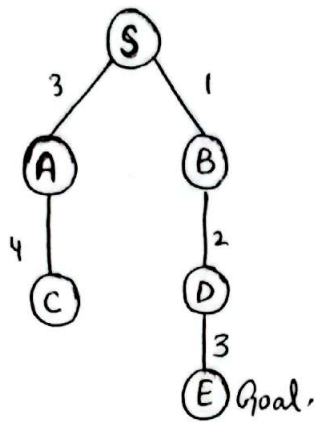
A	
B(2)	C(1)
B(2)	E(3)
E(3)	D(7)
D(7)	F(4)
D(7)	

Path:

$A \rightarrow C(1) \rightarrow B(2) \rightarrow E(3) \rightarrow F(4)$

Completeness: Complete

Uniform Cost Search



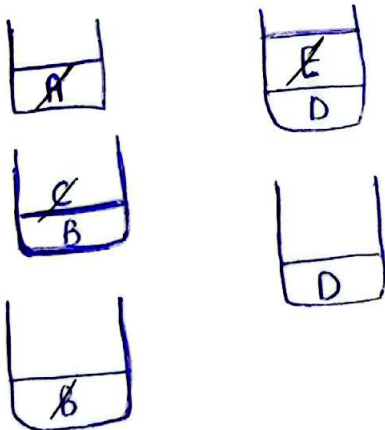
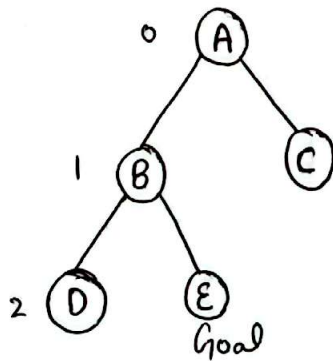
S	
A(3)	B(1)
A(3)	D(3)
D(3)	C(7)
C(7)	E(6)
C(7)	

Path:

$S \rightarrow B(1) \rightarrow A(3) \rightarrow D(3) \rightarrow E(6)$.

Completeness: Complete.

Q5: Depth-first Search



Path:

$A \rightarrow C \rightarrow B \rightarrow E$

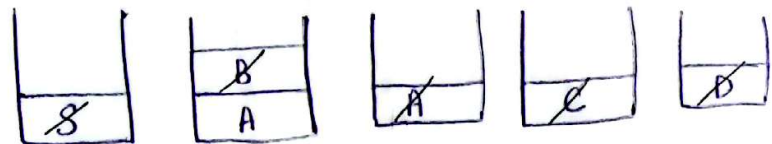
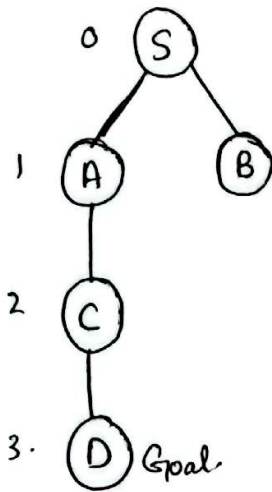
Time Complexity:

$$O(b^d) = 2^2$$

$$= 4$$

Completeness: Complete

Q6. Depth-first Search



Path:

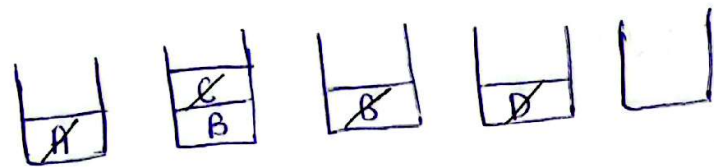
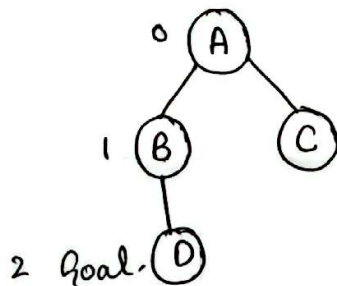
$S \rightarrow B \rightarrow A \rightarrow C \rightarrow D$

Completeness: Complete.

Time Complexity:

$$O(b^d) = 2^3 = 8.$$

Q7. Depth-Limited (Limit = 2).



Path:

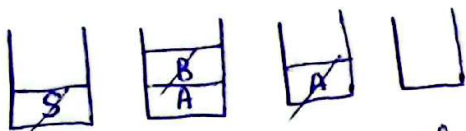
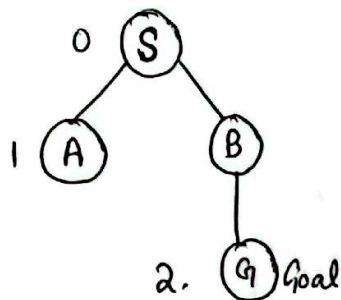
$A \rightarrow C \rightarrow B \rightarrow D$

Time Complexity:

$$O(b^l) = 2^2 = 4.$$

Completeness: Complete

Q8. Depth-Limited (Limit = 1)

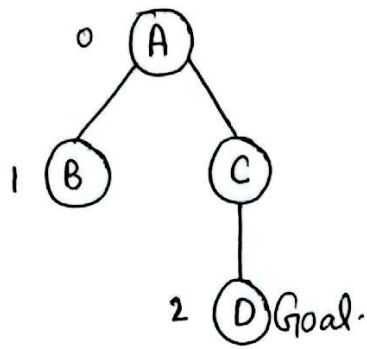


Path: No path because goal is not found.

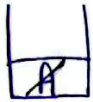
Completeness: Not Complete.

$$\text{Complexity} \times O(b^d) = 2^1 = 2.$$

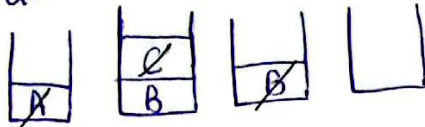
Iterative Deepening DFS



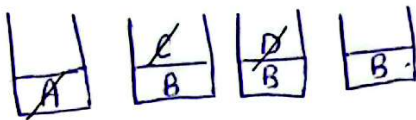
Itr 1



Itr 2



Itr 3



Path:

$A \rightarrow C \rightarrow D$

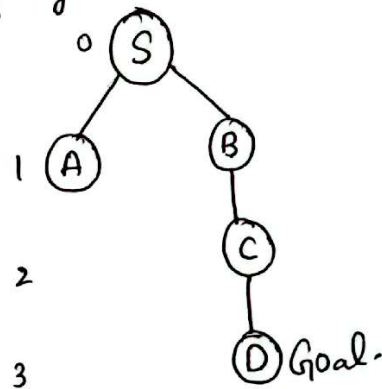
Time Complexity

$$O(b^l) = 2^2$$

$$= 4$$

Completeness: Complete

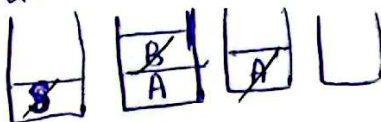
Q10: Iterative Deepening DFS



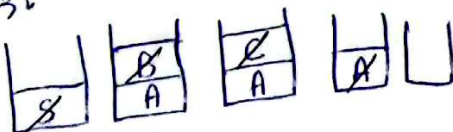
Itr 1



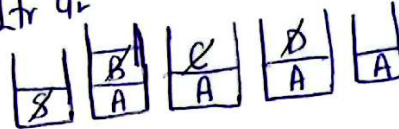
Itr 2



Itr 3



Itr 4



Path:

$S \rightarrow B \rightarrow C \rightarrow D$

Time Complexity

$$O(b^l) = 2^3$$

$$= 8$$

Completeness: Complete